

The Brain-Changing Benefits of Exercise | Wendy Suzuki | TED

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Tóm tắt

Nhà thần kinh học Wendy Suzuki kể lại cách bà vô tình phát hiện ra tác động biến đổi não bộ của việc tập thể dục khi tự thử nghiệm trên chính mình, từ đó chuyển hẳn hướng nghiên cứu sang chủ đề này. Bà giải thích tập thể dục mang lại lợi ích tức thời (tăng chất dẫn truyền thần kinh, cải thiện tâm trạng, sự chú ý và thời gian phản ứng) và lợi ích lâu dài (sinh tế bào não mới, tăng thể tích hồi hải mã và vỏ não trước trán, bảo vệ não khỏi suy giảm nhận thức và các bệnh thoái hóa thần kinh). Bà đưa ra khuyến nghị tối thiểu: tập aerobic 3-4 lần/tuần, mỗi lần 30 phút, và kết thúc bài nói bằng một phút vận động cùng khán giả.

Nội dung

[00:00] Mở đầu

What if I told you there was something that you can do right now that would have an immediate positive benefit for your brain, including your mood and your focus? And what if I told you that same thing could actually last a long time and protect your brain from different conditions like depression, Alzheimer's disease, or dementia — would you do it? Yes. I am talking about the powerful effects of physical activity — that is, simply moving your body — has immediate, long-lasting, and protective benefits for your brain, and that can last for the rest of your life.

So what I want to do today is tell you a story about how I used my deep understanding of neuroscience as a professor of neuroscience to essentially do an experiment on myself, in which I discovered the science underlying why exercise is the most transformative thing that you can do for your brain today.

[01:39] Vỏ não trước trán (Prefrontal Cortex)

Now, as a neuroscientist, I know that our brains — that is, the thing in our head right now — is the most complex structure known to humankind. But it's one thing to talk about the brain, and it's another to see it. So here is a real preserved human brain, and it's going to illustrate two key areas that we're going to talk about today.

The first is the prefrontal cortex, right behind your forehead, critical for things like decision-making, focus, attention, and your personality.

[02:07] Hồi hải mã (Hippocampus)

The second key area is located in the temporal lobe, shown right here. You have two temporal lobes in your brain, the right and the left, and deep in the temporal lobe is a key structure critical for your ability to form and retain new long-term memories for facts and events — and that structure is called the hippocampus.

So I have always been fascinated with the hippocampus. How could it be that an event that lasts just a moment — say, your first kiss, or the moment your first child was born — can form a memory that has changed your brain, that lasts an entire lifetime? That's what I want to understand. I wanted to start and record the activity of individual brain cells in the hippocampus as subjects were forming new memories, and essentially trying to decode how those brief bursts of electrical activity — which is how neurons communicate with each other — how those brief bursts either allowed us to form a new memory, or did not.

[03:13] Tác động thay đổi não bộ của việc tập thể dục

But a few years ago, I did something very unusual in science: as a full professor of neuroscience, I decided to completely switch my research program, because I encountered something that was so amazing, with the potential to change so many lives, that I had to study it. I discovered — and I experienced — the brain-changing effects of exercise, and I did it in a completely inadvertent way.

I was actually at the height of all the memory work that I was doing. Data was pouring in, I was becoming known in my field for all of this memory work, and it should have been going great — scientifically, it was — but when I stuck my head out of my lab door, I noticed something: I had no social life. I spent too much time listening to those brain cells in a dark room by myself. I didn't move my body at all. I had gained 25 pounds, and it actually took me many years to realize it was actually miserable, and I shouldn't be miserable.

I went on a river rafting trip by myself, because I had no social life, and I came back thinking, "Oh my god, I was the weakest person on that trip." And I came back with a mission. I said, "I'm never gonna feel like the weakest person on a river rafting trip again." And that's what made me go to the gym. I focused my type-A personality on going to all the exercise classes

at the gym. I tried everything — kickboxing, dance, yoga, a step class — and at first it was really hard, but what I noticed is that after every sweat-inducing workout that I tried, I had this great mood boost and this great energy boost, and that's what kept me going back to the gym. I started feeling stronger, I started feeling better, I even lost that 25 pounds. Now, fast forward a year and a half into this regular exercise program, and I noticed something that really made me sit up and take notice. I was sitting at my desk writing a research grant, and a thought went through my mind that had never gone through my mind before, and that thought was, "Gee, grant writing is going well today." All the scientists always laugh when I say that, because grant writing never goes well — it's so hard, you're always pulling your hair out trying to come up with that million-dollar winning idea.

But I realized that the grant writing was going well because I was able to focus and maintain my attention for longer than I had before, and my long-term memory for what I was studying in my own lab seemed to be better in me. And that's when I put it together: maybe all that exercise that I had included and added to my life was changing my brain. Maybe I did an experiment on myself without even knowing it.

So, as a curious neuroscientist, I went to the literature to see what I could find about what we knew about the effects of exercise on the brain, and what I found was an exciting and a growing literature that was essentially showing everything that I noticed in myself: better mood, better energy, better memory, better attention. And the more I learned, the more I realized how powerful exercise was, which eventually led me to the big decision to completely shift my research focus.

[06:35] Tập thể dục là điều biến đổi mạnh mẽ nhất bạn có thể làm cho não bộ

And so, now, after several years of really focusing on this question, I've come to the following conclusion: that exercise is the most transformative thing that you can do for your brain today, for the following three reasons.

Number one, it has immediate effects on your brain. A single workout that you do will immediately increase levels of neurotransmitters like dopamine, serotonin, and noradrenaline. That is going to increase your mood right after that workout — exactly what I was feeling. My lab showed that a single workout can improve your ability to shift and focus attention, and that focus improvement will last for at least two hours. And finally, studies have shown that a single workout will improve your reaction times, which basically means that you are going to be faster at catching that cup of Starbucks that falls off the counter — which is very, very important.

But these immediate effects are transient — they help you right after. What you have to do is do what I did: that is, change your exercise regime, increase your cardio-respiratory function, to get the long-lasting effects. And these effects are long-lasting because exercise actually changes the brain's anatomy, physiology, and function.

[07:49] Hồi hải mã và những thay đổi lâu dài

Let's start with my favorite brain area, the hippocampus. With exercise, the hippocampus actually produces brand-new brain cells — new brain cells in the hippocampus that actually increase its volume, as well as improve your long-term memory. And that includes in you and me.

[08:16] Chức năng chú ý (Attention Function)

Number two, the most common finding in neuroscience studies looking at the effects of long-term exercise is improved attention function, dependent on your prefrontal cortex. You not only get better focus and attention, but the volume of the hippocampus increases as well. And finally, you not only get immediate effects of mood with exercise, but those last for a long time, so you get long-lasting increases in those good-mood neurotransmitters.

But really, the most transformative thing that exercise will do is its protective effects on your brain. Here, you can think about the brain like a muscle: the more you're working out, the bigger and stronger your hippocampus and prefrontal cortex gets. Why is that important? Because the prefrontal cortex and the hippocampus are the two areas that are most susceptible to neurodegenerative diseases and normal cognitive decline in aging. So with increased exercise over your lifetime, you're not going to cure dementia or Alzheimer's disease, but what you're going to do is create the strongest, biggest hippocampus and prefrontal cortex, so it takes longer for these diseases to actually have an effect. You can think of exercise, therefore, as a supercharged 401(k) for your brain — and it's even better because it's free.

[09:48] Lượng tập thể dục tối thiểu cần thiết

So this is a point in the talk where everybody says, "That sounds so interesting, Wendy, but I really only want to know one thing, and that is: just tell me the minimum amount of exercise I need to get all these changes." So I'm gonna tell you the answer to that question first.

Good news: you don't have to become a triathlete to get these effects. The rule of thumb is you want to get three to four times a week of exercise, minimum 30 minutes per exercise session, and you want to get aerobic exercise in — that is, get your heart rate up. And the good news is you don't have to go to the gym or get a very expensive gym membership: add an extra walk around the block, do a power walk, take the stairs, and power vacuuming can be as good as the aerobics class that you were going to take at the gym.

So I have gone from memory pioneer to exercise explorer, from going into the innermost workings of the brain to try to understand how exercise can improve our brain function. And my goal in my lab right now is to go beyond that rule of thumb that I just gave you — three to four times a week, 30 minutes — I want to understand the optimum exercise prescription for

you, at your age, at your fitness level, for your genetic background, to maximize the effects of exercise today and also to improve your brain and protect your brain the best for the rest of your life.

But it's one thing to talk about exercise, and it's another to do it. So I'm going to invoke my power as a certified exercise instructor to ask you all to stand up. We're gonna do just one minute of exercise, call and response — just do what I do, say what I say, and make sure you don't punch your neighbor, okay?

(Music, exercise demonstration: "I am strong," "I am Wonder Woman strong," "I am inspired," "I am on fire.")

I want to leave you with one last thought, and that is: bringing exercise into your life will not only give you a happier, more productive life today, but it will protect your brain from incurable diseases, and in this way it will change the trajectory of your life for the better. Thank you very much.

Điểm chính

- Tập thể dục có lợi ích tức thời: tăng dopamine, serotonin, noradrenaline, cải thiện tâm trạng ngay sau buổi tập.
 - Một buổi tập duy nhất cải thiện khả năng chuyển đổi và tập trung sự chú ý, hiệu ứng kéo dài ít nhất 2 giờ.
 - Một buổi tập duy nhất cũng cải thiện thời gian phản ứng.
 - Các hiệu ứng tức thời chỉ mang tính tạm thời — cần duy trì tập luyện đều đặn để có hiệu ứng lâu dài.
 - Tập luyện lâu dài làm thay đổi giải phẫu, sinh lý và chức năng của não: sinh tế bào não mới ở hồi hải mã, tăng thể tích hồi hải mã, cải thiện trí nhớ dài hạn.
 - Tập luyện lâu dài cải thiện chức năng chú ý phụ thuộc vào vỏ não trước trán.
 - Tác dụng bảo vệ quan trọng nhất: hồi hải mã và vỏ não trước trán to và khỏe hơn giúp làm chậm tác động của các bệnh thoái hóa thần kinh và suy giảm nhận thức do tuổi tác (được ví như "quỹ hưu trí 401k siêu cấp cho não bộ").
 - Khuyến nghị tối thiểu: tập aerobic 3-4 lần/tuần, mỗi lần tối thiểu 30 phút, không cần phòng gym đắt tiền (đi bộ nhanh, leo cầu thang, hút bụi mạnh cũng có tác dụng tương đương).
 - Câu chuyện cá nhân: diễn giả từng tăng 25 pound, không vận động, sau đó tập thể dục đều đặn giúp cải thiện tâm trạng, năng lượng, khả năng tập trung khi viết đề xuất tài trợ nghiên cứu — từ đó chuyển hẳn hướng nghiên cứu.
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Ghi chú

- Transcript được tạo tự động (auto-generated captions), có thể có lỗi nhỏ về chính tả hoặc nhận dạng giọng nói.
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